

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Total Marks: 50

QUESTIONS: 5

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

(192521331)

I Mexlin Epzibah (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

S. Mexlin
Signature of the student:

86.11
48

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately 2	Understands problem with minor gaps 1.5	Partial understanding; misses key elements 1	Misinterprets or unable to understand problem 0
Method Selection	20%	Selects most appropriate method/formula with strong justification 2	Selects correct method with minor errors 1.5	Inappropriate or partially correct method 1	Unable to select method 0
Solution Process	25%	Logical, systematic steps with correct approach throughout 2.5	Mostly correct steps with minor mistakes 2	Disorganized steps; multiple errors 1.5	No clear process 0
Accuracy of Calculation	20%	All calculations accurate; correct final answer 2	Minor calculation errors 1.5	Frequent errors; incorrect final answer 1	No valid calculations 0
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance 1.5	Basic interpretation with minor omissions 1	Limited or unclear interpretation 0.5	No interpretation 0

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Problem (20%)
understanding

Method (20%)
selection

Solution
(25%)

Accuracy
(20%)

Results
(15%)

Q1

2

1.5

2.5

2

1.5 9.5

Q2

1.5

1.5

2

2

1 8

Q3

1.5

2

2.5

1.5

1 8.5

Q4

2

1.5

2

2

1 8.5

Q5

2

2

1.5

2

1 8.5

1. Subnetting a /24 Network for Three Labs

Question: A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three labs using fixed subnet masks — Lab A (/26), Lab B (/27), and Lab C (/28). Determine the network address, valid host range, broadcast address, and number of usable hosts for each lab.

Subnetting is the process of dividing a large network block into smaller logical sub-networks by borrowing bits from the host portion of the address for the network portion. Since the labs are given different fixed prefix lengths, this is an example of Variable Length Subnet Masking (VLSM), where each subnet is sized according to its own requirement rather than all subnets being of equal size. To avoid overlap, the subnets are allocated one after another, starting from the base network address 192.168.20.0, with the largest subnet (Lab A) placed first.

Calculation:

Step 1: Lab A uses a /26 mask (255.255.255.192). The number of host bits is $32 - 26 = 6$, so block size = $2^6 = 64$. Lab A therefore occupies 192.168.20.0 to 192.168.20.63. The first address (192.168.20.0) is the network address and the last address (192.168.20.63) is the broadcast address, leaving 62 usable host addresses ($2^6 - 2 = 62$).

Step 2: Lab B uses a /27 mask (255.255.255.224). Host bits = $32 - 27 = 5$, block size = $2^5 = 32$. Since Lab A already used up to .63, Lab B starts at the next available boundary, 192.168.20.64, and extends to 192.168.20.95. This gives $2^5 - 2 = 30$ usable hosts.

Step 3: Lab C uses a /28 mask (255.255.255.240). Host bits = $32 - 28 = 4$, block size = $2^4 = 16$. Lab C starts right after Lab B, at 192.168.20.96, and extends to 192.168.20.111, giving $2^4 - 2 = 14$ usable hosts.

4: The remaining addresses, 192.168.20.112 to 192.168.20.255 (144 addresses), stay unassigned and can be reserved for future labs or expansion.

Summary Table:

Lab	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Lab A (/26)	255.255.255.192	192.168.20.0	192.168.20.1 – .62	192.168.20.63	62
Lab B (/27)	255.255.255.224	192.168.20.64	192.168.20.65 – .94	192.168.20.95	30
Lab C (/28)	255.255.255.240	192.168.20.96	192.168.20.97 – .110	192.168.20.111	14

Subnetting of 192.168.20.0/24 into Lab A, Lab B, Lab C

Total block: 192.168.20.0/24 (256 addresses)

Lab A (/26)	Lab B (/27)	Lab C (/28)	Unused
192.168.20.0 - .63 64 addresses	192.168.20.64 - .95 32 addresses	192.168.20.96 - .111 16 addresses	192.168.20.112 - .255 144 addresses

Conclusion

By applying VLSM to the /24 block, the administrator successfully creates three right-sized subnets for the labs — 62 hosts for Lab A, 30 hosts for Lab B, and 14 hosts for Lab C — while still keeping 144 addresses in reserve for future departments, demonstrating efficient utilisation of the given address space

2.Verifying Two Systems in the Same Subnet

Question: A network administrator has configured two systems: System 1 with IP address 192.168.1.130 and subnet mask 255.255.255.192, and System 2 with IP address 192.168.1.190 and the same subnet mask. Determine the subnet range, network address, and broadcast address, and check whether both IP addresses belong to the same subnet.

Concept:

The subnet mask 255.255.255.192 corresponds to a /26 prefix, since 192 in binary is 11000000, i.e. 2 network bits borrowed from the last octet, giving 26 network bits in total. This means the host portion has 6 bits, producing a block size of $2^6 = 64$ addresses per subnet. To find which subnet an address belongs to, the address is compared against successive multiples of the block size (0, 64, 128, 192) within the final octet.

Step-by-Step Calculation

Step 1: Block size = $256 - 192 = 64$. The subnet boundaries in the last octet are therefore 0, 64, 128, and 192.

Step 2: For System 1 (192.168.1.130): 130 lies between 128 and 191, so System 1 belongs to the subnet 192.168.1.128/26.

Step 3: For System 2 (192.168.1.190): 190 also lies between 128 and 191, so System 2 belongs to the same subnet 192.168.1.128/26.

Step 4: For subnet 192.168.1.128/26: Network address = 192.168.1.128, Broadcast address = 192.168.1.191 (the last address before the next boundary at 192), and Valid host range = 192.168.1.129 to 192.168.1.190.

Step 5: Checking membership: System 1 (.130) and System 2 (.190) both fall strictly within 129–190, so both are valid, usable host addresses in the same subnet. In fact, System 2 (.190) is the last usable address in that subnet, just one below the broadcast address.

Summary Table:

Parameter	Value
Subnet Mask	255.255.255.192 (/26)

Parameter	Value
Network Address	192.168.1.128
Valid Host Range	192.168.1.129 – 192.168.1.190
Broadcast Address	192.168.1.191
Usable Hosts	62 ($2^6 - 2$)
System 1 (192.168.1.130)	Valid host — within range
System 2 (192.168.1.190)	Valid host — within range (last usable address)
Same Subnet?	Yes — both systems can communicate directly without a router

Subnet 192.168.1.128/26 containing System 1 and System 2

Subnet mask 255.255.255.192 (/26) - Block size 64

Net Addr	System 1	Usable Host Range	System 2	Broadcast
192.168.1.128	192.168.1.130	192.168.1.129 - 190	192.168.1.190	192.168.1.191

Conclusion:

Since both 192.168.1.130 and 192.168.1.190 fall within the host range 192.168.1.129–192.168.1.190 of the same /26 subnet (192.168.1.128), System 1 and System 2 reside in the same broadcast domain. They can exchange traffic directly at Layer 2 without requiring a router or Layer 3 gateway.

VLSM Allocation for a /16 Organisation Network

Question: An organisation has been allocated the IP block 172.16.0.0/16. The administrator has decided subnet masks for four departments: HR (/25), Sales (/26), IT (/27), and Admin (/28). Determine the network address, valid host range, and broadcast address for each subnet, the number of usable hosts per department, and the remaining unused address range.

Each department is given a different prefix length, this is again a VLSM allocation. The general rule followed is to allocate subnets in decreasing order of size — the largest subnet first — so that each smaller block can be carved out from the remaining contiguous space without wasting addresses or causing overlaps. All four departments are allotted from the very start of the /16 block, 172.16.0.0.

Calculation:

Step 1: HR (/25): Host bits = $32 - 25 = 7$, block size = $2^7 = 128$. HR occupies 172.16.0.0 to 172.16.0.127. Network = 172.16.0.0, Broadcast = 172.16.0.127, Usable hosts = $2^7 - 2 = 126$.

Step 2: Sales (/26): Host bits = 6, block size = 64. Starting right after HR, Sales occupies 172.16.0.128 to 172.16.0.191. Network = 172.16.0.128, Broadcast = 172.16.0.191, Usable hosts = $2^6 - 2 = 62$.

Step 3: IT (/27): Host bits = 5, block size = 32. IT starts at 172.16.0.192 and occupies up to 172.16.0.223. Network = 172.16.0.192, Broadcast = 172.16.0.223, Usable hosts = $2^5 - 2 = 30$.

Step 4: Admin (/28): Host bits = 4, block size = 16. Admin starts at 172.16.0.224 and occupies up to 172.16.0.239. Network = 172.16.0.224, Broadcast = 172.16.0.239, Usable hosts = $2^4 - 2 = 14$.

Step 5: Total addresses consumed = $128 + 64 + 32 + 16 = 240$, out of the 65,536 addresses available in the /16 block. This leaves the range 172.16.0.240 to 172.16.255.255 completely unused and available for future departments or growth.

Summary Table

Dept.	Mask	Network Address	Valid Host Range	Broadcast	Usable Hosts
HR (/25)	255.255.255.128	172.16.0.0	172.16.0.1 – .126	172.16.0.127	126
Sales (/26)	255.255.255.192	172.16.0.128	172.16.0.129 – .190	172.16.0.191	62
IT (/27)	255.255.255.224	172.16.0.192	172.16.0.193 – .222	172.16.0.223	30
Admin (/28)	255.255.255.240	172.16.0.224	172.16.0.225 – .238	172.16.0.239	14

Subnetting of 172.16.0.0/16 into HR, Sales, IT, Admin

Total block: 172.16.0.0/16 (65,536 addresses)

HR (/25)	Sales (/26)	IT (/27)	Admin (/28)	Unused / Reserved for future use
172.16.0.0-127 126 hosts	172.16.0.128-191 62 hosts	172.16.0.192-223 30 hosts	172.16.0.224-239 16 hosts	172.16.0.240-172.16.255.255

Remaining Unused Range

After allocating all four departments, the unused address space is 172.16.0.240 – 172.16.255.255, which is far larger than everything allotted so far. This remaining space can be subnetted again in the future using the same VLSM approach whenever new departments are added.

Conclusion

... scheme allocates exactly the number of hosts each department needs — 126, 62, 30, respectively — while consuming only a small fraction (240 out of 65,536 addresses) of total /16 block, leaving abundant room for future expansion.

4. Fixed-Length Subnetting of a /24 Block into /27 Subnets

Question: A company uses the IP address 172.16.10.0/24 and subnets it further using a /27 mask for its internal departments. Calculate the total number of subnets created and usable hosts per subnet, determine which subnet 172.16.10.78 belongs to along with its network and broadcast addresses, and check whether 172.16.10.95 lies in the same subnet.

Concept

This is Fixed Length Subnet Masking (FLSM), where the original /24 network is divided into equal-sized subnets by extending the mask uniformly to /27. Since 3 extra bits are borrowed ($27 - 24 = 3$), the number of subnets created is $2^3 = 8$, and each subnet has $32 - 27 = 5$ host bits, giving a block size of $2^5 = 32$ addresses per subnet.

Step-by-Step Calculation

Step 1: Number of subnets = $2^{(\text{borrowed bits})} = 2^3 = 8$ subnets.

Step 2: Usable hosts per subnet = $2^{(\text{host bits})} - 2 = 2^5 - 2 = 30$ hosts.

Step 3: Block size = 32, so the subnet boundaries (last octet) are: 0, 32, 64, 96, 128, 160, 192, 224 — these mark the 8 subnets, each 32 addresses wide.

Step 4: Locating 172.16.10.78: it lies between the boundaries 64 and 95, so it falls in the third subnet, 172.16.10.64/27.

Step 5: For subnet 172.16.10.64/27: Network address = 172.16.10.64, Broadcast address = 172.16.10.95 (one less than the next boundary, 96), and Valid host range = 172.16.10.65 to 172.16.10.94.

Checking 172.16.10.95: this address equals the broadcast address of the 172.16.10.64/27 subnet calculated above. It is technically inside the same subnet block, but it is reserved for broadcast traffic — it is not a usable host address and cannot be assigned to any device.

Summary Table

Parameter	Value
Subnet Mask	255.255.255.224 (/27)
Total Subnets Created	8
Usable Hosts per Subnet	30
Subnet containing .78	172.16.10.64/27 (3rd subnet)
Network Address	172.16.10.64
Valid Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95
Is 172.16.10.95 usable?	No — it is the broadcast address of this subnet

8 Subnets of 172.16.10.0/24 using /27 mask (Subnet 3 highlighted contains .78)

Block size = 32 addresses per subnet | 172.16.10.78 falls in Subnet 3: 172.16.10.64/27

Subnet 1	Subnet 2	Subnet 3	Subnet 4	Subnet 5	Subnet 6	Subnet 7	Subnet 8
.0 - .31	.32 - .63	.64 - .95	.96 - .127	.128 - .159	.160 - .191	.192 - .223	.224 - .255

Conclusion

Dividing 172.16.10.0/24 with a /27 mask produces 8 equal subnets of 30 usable hosts each. The address 172.16.10.78 belongs to the 172.16.10.64/27 subnet, and although 172.16.10.95 falls

same block, it serves as the subnet's broadcast address rather than an assignable host

IPv6 Address Compression, Expansion, and Prefix

Question: A network uses the IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Compress this address into its shortest form, expand the compressed form back to full notation, determine the network prefix using a /64 prefix length, and calculate the total number of possible host addresses in that network.

Concept

An IPv6 address is 128 bits long, written as eight groups of four hexadecimal digits separated by colons. Two compression rules are applied to shorten it: (1) leading zeros within each group may be omitted, and (2) one single contiguous run of all-zero groups may be replaced with a double colon '::', but this shorthand can be used only once in an address to avoid ambiguity about how many zero groups it represents. A /64 prefix is the most common allocation size, splitting the address exactly into a 64-bit network prefix and a 64-bit interface identifier.

Step-by-Step Working

Step 1: Start with the full address: 2001:0db8:0000:0000:0000:ff00:0042:8329.

Step 2: Remove leading zeros in each group: 2001:db8:0:0:0:ff00:42:8329.

Step 3: Replace the longest run of consecutive all-zero groups (the three consecutive '0' groups) with '::': 2001:db8::ff00:42:8329. This is the compressed form.

Step 4: To expand it back: replace '::' with enough all-zero groups to restore all 8 groups, and pad each group back to 4 hex digits: 2001:0db8:0000:0000:0000:ff00:0042:8329 — matching the original address exactly.

Step 5: Applying a /64 prefix means the first 64 bits (first four groups) form the network prefix and the last 64 bits (last four groups) form the interface ID. Network prefix = 2001:0db8:0000:0000:0000:0000:0000/64, written in compressed form as 2001:db8::/64.

Total host addresses possible in a /64 network = $2^{(128 - 64)} = 2^{64} = 18,446,744,073,709,551,616$ addresses — an extremely large address space reserved for a single subnet, unlike IPv4 where such a size would be inconceivable.

Summary Table

Parameter	Value
Full (Expanded) Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed Address	2001:db8::ff00:42:8329
Prefix Length Used	/64
Network Prefix	2001:db8::/64
Interface ID Portion	0000:ff00:0042:8329 (last 64 bits)
Total Possible Addresses	$2^{64} = 18,446,744,073,709,551,616$

IPv6 Address Structure: 2001:0db8:0000:0000:0000:ff00:0042:8329 /64

Network Prefix (first 64 bits)					Interface ID (last 64 bits)		
2001	0db8	0000	0000	0000	ff00	0042	8329

Conclusion:

The address 2001:0db8:0000:0000:0000:ff00:0042:8329 compresses to 2001:db8::ff00:42:8329 by dropping leading zeros and collapsing the single run of zero groups, and expands back identically by reversing both steps. Under a /64 prefix, the network portion is 2001:db8::/64, leaving a 64-bit interface ID capable of addressing over 18 quintillion hosts on that one subnet.